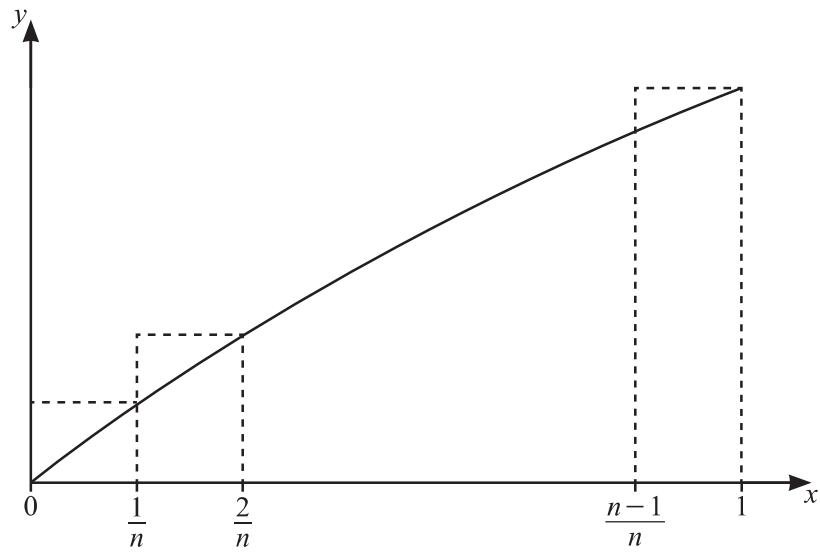




The diagram shows the curve with equation $y = \ln(1+x)$ for $0 \leq x \leq 1$, together with a set of n rectangles each of width $\frac{1}{n}$.

(a) By considering the sum of the areas of these rectangles, show that $\int_0^1 \ln(1+x) dx < U_n$, where

$$U_n = \frac{1}{n} \ln \frac{(2n)!}{n!} - \ln n. \quad [4]$$

[illegible]

- (b) Use a similar method to find, in terms of n , a lower bound L_n for $\int_0^1 \ln(1+x) dx$. [4]

This image shows a full page of a worksheet designed for handwriting practice. It features 18 evenly spaced, horizontal dashed lines that run across the entire width of the page. The background is plain white, providing a clear space for writing. There are no margins, text, or other markings on the page.

- (c) By simplifying $U_n - L_n$, show that $\lim_{n \rightarrow \infty} (U_n - L_n) = 0$. [2]

[illegible]